(a) Corpus construction pipeline

1 · Source admission
whitelisted journals
variety & rootstock records (UCR entry list, citrus monograph)
authorized private data

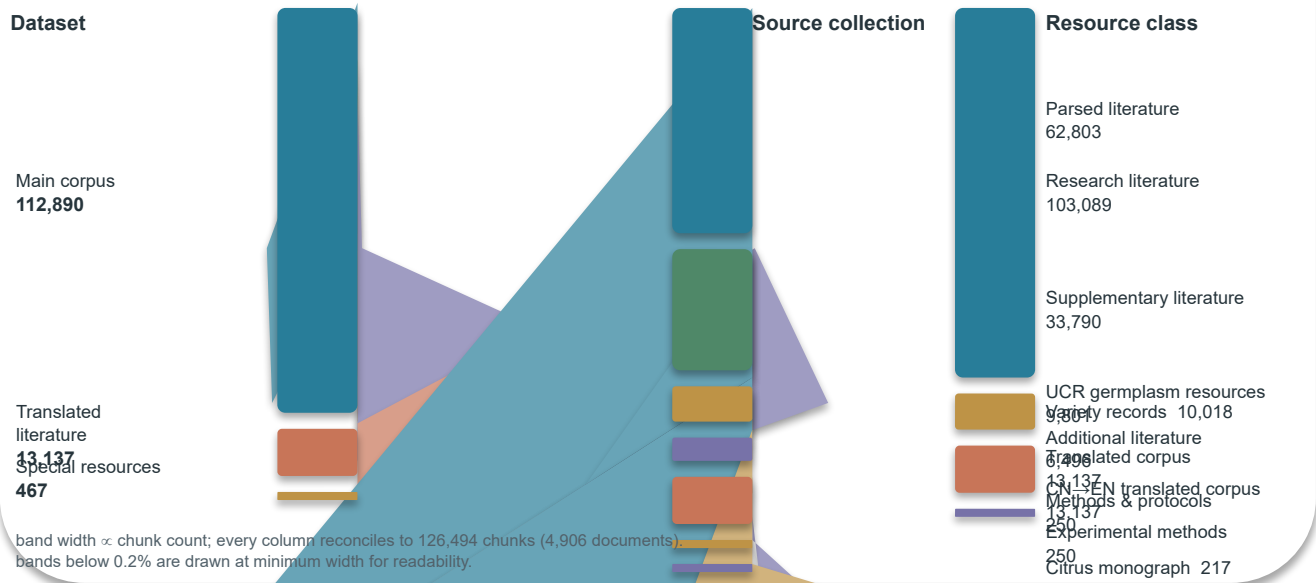
2 · Cleaning
encoding normalization
layout / noise removal
content + normalized-text dedup (SHA-256)
one representative copy

3 · Metadata QA
title / year / DOI rules
format & range checks
manual exception review

excluded / flagged:
unparseable · too short

4 · Chunks & index
heading-based segmentation
recursive chunks (target ≈ 800 chars)
dense e5-large · LanceDB
sparse BM25Plus
frozen snapshot+manifest

(b) Dataset, source collection and resource class (chunks)



band width ∝ chunk count; every column reconciles to 126,494 chunks (4,906 documents).
bands below 0.2% are drawn at minimum width for readability.